

The empirical recurrence for OEIS A297594: recovering a conjecture that is not written down

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Abstract

OEIS A297594 records “Empirical recurrence of order 83”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 134$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 83, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 83 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A297594 is “Number of $n \times 7$ 0..1 arrays with every 1 horizontally, diagonally or antidiagonally adjacent to 1 or 5 neighboring 1s.”. It has offset 1 and begins

13, 121, 1141, 10131, 103673, 975485, 9403199, 90862063, ...

The entry states:

Empirical recurrence of order 83 (see link above)

As of the “Last modified” line on the live entry (Jan 01 2018 12:07:30, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 16384$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 134$ of the 16384, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 134$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 268$ exact terms of the model returns order 83 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{83} c_i a(n-i),$$

c_1	3	c_{22}	85135191137	c_{43}	−185775251396977	c_{64}	−2781983556156
c_2	27	c_{23}	−449677488145	c_{44}	−83186976962210	c_{65}	−4470579590650
c_3	417	c_{24}	204379070970	c_{45}	43047722494437	c_{66}	−1665804703634
c_4	414	c_{25}	−848306116327	c_{46}	197637151894261	c_{67}	536067596054
c_5	−4600	c_{26}	3780012497967	c_{47}	−47744919259599	c_{68}	1552466157224
c_6	−46133	c_{27}	−2284089667550	c_{48}	−75781999485159	c_{69}	446421525480
c_7	−77861	c_{28}	3423094080944	c_{49}	−55687617040074	c_{70}	227063524940
c_8	317334	c_{29}	−16236305938059	c_{50}	51711286626852	c_{71}	−225074103764
c_9	1798244	c_{30}	3432018639329	c_{51}	63852395333892	c_{72}	−50834042656
c_{10}	3233513	c_{31}	6353183480815	c_{52}	−9350369367434	c_{73}	−20992112528
c_{11}	−9931732	c_{32}	37354980474487	c_{53}	−30065698173018	c_{74}	24415895368
c_{12}	−19319994	c_{33}	4961782592309	c_{54}	−54854115924204	c_{75}	−3221774032
c_{13}	−92627283	c_{34}	−66469131208881	c_{55}	19337359010076	c_{76}	−459389288
c_{14}	255197142	c_{35}	−46173713017986	c_{56}	22974281518276	c_{77}	−1643160288
c_{15}	57484398	c_{36}	23635358789388	c_{57}	19234317990120	c_{78}	338701408
c_{16}	1736724333	c_{37}	161967281744089	c_{58}	−20908015862208	c_{79}	150139904
c_{17}	−6081074602	c_{38}	−36815330025728	c_{59}	−20461720507755	c_{80}	−40194272
c_{18}	2377403659	c_{39}	−136234865794588	c_{60}	−6100981802314	c_{81}	−10979680
c_{19}	−18831226892	c_{40}	−57069418562484	c_{61}	9630435052730	c_{82}	3603456
c_{20}	55184517456	c_{41}	165936432008661	c_{62}	11611272149656	c_{83}	−73728
c_{21}	−6260568727	c_{42}	109349333115893	c_{63}	4786977934012		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 83, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $83 + 4$ terms removed returns order 83 as well, so $\deg q_{\min} = 83 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 18 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $83 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 83 that this sequence satisfies — and that family turns out to have exactly one member.

References

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